

Data Dictionary — Healthcare Data Warehouse

DEPI Microsoft Data Engineer Track — Milestone 2

Schema: dbo (Azure Synapse Analytics Dedicated SQL Pool)

dim_patient

Type: Dimension — SCD Type 2

Distribution: REPLICATE

Grain: One row per patient per version (insurance/address/marital status change)

Source: Synthea CSV → Bronze → Silver → Gold

SCD-Tracked Columns: insurance_type, city, state, marital_status

Column	Data Type	Nullable	Description	Valid Values / Example
patient_key	INT	NO	Surrogate primary key. Unique per SCD version.	1, 2, 3...
patient_id	NVARCHAR(50)	NO	Natural key — Synthea UUID. Stable across versions.	"a1b2c3d4-..."
first_name	NVARCHAR(100)	NO	Patient first name (synthetic).	"Maria"
last_name	NVARCHAR(100)	NO	Patient last name (synthetic).	"Gonzalez"
birth_date	DATE	NO	Date of birth. Used to compute age.	1975-04-22
gender	NVARCHAR(10)	NO	Patient gender.	M / F / Other
race	NVARCHAR(50)	YES	Patient race as coded by Synthea.	"White", "Black", "Asian"
ethnicity	NVARCHAR(50)	YES	Patient ethnicity.	"Hispanic", "Non-Hispanic"
city	NVARCHAR(100)	YES	Patient city of residence. SCD-tracked.	"Cairo"
state	NVARCHAR(50)	YES	Patient state/governorate. SCD-tracked.	"Giza"
zip_code	NVARCHAR(20)	YES	Postal code.	"12345"
insurance_type	NVARCHAR(50)	YES	Current insurance coverage. SCD-tracked.	private / medicaid / medicare / self-pay

Column	Data Type	Nullable	Description	Valid Values / Example
marital_status	NVARCHAR(20)	YES	Marital status. SCD-tracked.	Single / Married / Divorced / Widowed
valid_from	DATE	NO	Date this record version became active.	2023-01-15
valid_to	DATE	YES	Date this record version expired. NULL = current.	2024-06-30 or NULL
is_current	BIT	NO	1 = active record for this patient. 0 = historical.	0 / 1
age_at_load	Computed	YES	Age in years as of query time.	48
age_group	Computed	YES	Banded age group.	0-17 / 18-34 / 35-49 / 50-64 / 65+
row_hash	NVARCHAR(64)	YES	SHA-256 of SCD-tracked columns. Used for change detection.	"3a4b5c..."

dim_provider

Type: Dimension — SCD Type 2

Distribution: REPLICATE

Grain: One row per provider per version (specialty/organization change)

Source: Flask REST API → Bronze → Silver → Gold

SCD-Tracked Columns: specialty, sub_specialty, organization

Column	Data Type	Nullable	Description	Valid Values / Example
provider_key	INT	NO	Surrogate primary key.	1, 2, 3...
provider_id	NVARCHAR(50)	NO	Natural key — provider UUID.	"p001-..."
provider_name	NVARCHAR(200)	NO	Full name of provider.	"Dr. Ahmed Hassan"
specialty	NVARCHAR(100)	YES	Medical specialty. SCD-tracked.	"Cardiology", "Orthopedics"
sub_specialty	NVARCHAR(100)	YES	Sub-specialty if applicable. SCD-tracked.	"Interventional Cardiology"
organization	NVARCHAR(200)	YES	Hospital/clinic affiliation. SCD-tracked.	"Cairo University Hospital"
city	NVARCHAR(100)	YES	City of practice.	"Alexandria"
state	NVARCHAR(50)	YES	State/governorate.	"Alexandria"
npi_number	NVARCHAR(20)	YES	National Provider Identifier (simulated).	"1234567890"

Column	Data Type	Nullable	Description	Valid Values / Example
valid_from	DATE	NO	Version activation date.	2022-07-01
valid_to	DATE	YES	Version expiry date. NULL = current.	NULL
is_current	BIT	NO	1 = current active version.	1
row_hash	NVARCHAR(64)	YES	SHA-256 hash for change detection.	"7f8e9a..."

dim_date

Type: Dimension — Static (Type 0)

Distribution: REPLICATE

Grain: One row per calendar day (2010-01-01 to 2035-12-31)

Source: Generated via recursive CTE at warehouse setup

Column	Data Type	Nullable	Description	Example
date_key	INT	NO	Surrogate PK in YYYYMMDD format.	20240315
full_date	DATE	NO	Calendar date.	2024-03-15
year	SMALLINT	NO	Calendar year.	2024
quarter	TINYINT	NO	Calendar quarter (1-4).	1
quarter_name	NVARCHAR(10)	NO	Quarter label.	"Q1"
month	TINYINT	NO	Month number (1-12).	3
month_name	NVARCHAR(15)	NO	Full month name.	"March"
month_short	NVARCHAR(5)	NO	Abbreviated month name.	"Mar"
week_of_year	TINYINT	NO	Week number within calendar year.	11
day_of_month	TINYINT	NO	Day number within month (1-31).	15
day_of_week	TINYINT	NO	Day of week (1=Sunday, 7=Saturday).	6
day_name	NVARCHAR(15)	NO	Full day name.	"Friday"
is_weekend	BIT	NO	1 if Saturday or Sunday.	1
is_holiday	BIT	NO	1 if public holiday (updated separately).	0
holiday_name	NVARCHAR(100)	YES	Name of holiday if applicable.	"National Day"
fiscal_year	SMALLINT	NO	Fiscal year (starts July 1).	2024
fiscal_quarter	TINYINT	NO	Fiscal quarter (1-4).	3
fiscal_month	TINYINT	NO	Month within fiscal year (1-12).	9

Column	Data Type	Nullable	Description	Example
days_in_month	TINYINT	NO	Total days in this calendar month.	31
is_last_day_of_month	BIT	NO	1 if this is the last day of the month.	0
iso_week	TINYINT	NO	ISO 8601 week number.	11
year_month	NVARCHAR(7)	NO	Year-Month label for grouping.	"2024-03"

dim_time

Type: Dimension — Static (Type 0)

Distribution: REPLICATE

Grain: One row per minute of the day (1,440 rows)

Source: Generated via CROSS JOIN at warehouse setup

Column	Data Type	Nullable	Description	Example
time_key	INT	NO	Surrogate PK in HHMM format.	1430
hour	TINYINT	NO	Hour of day (0-23).	14
minute	TINYINT	NO	Minute (0-59).	30
hour_minute	NVARCHAR(5)	NO	Formatted time string.	"14:30"
am_pm	NVARCHAR(2)	NO	AM or PM indicator.	"PM"
shift	NVARCHAR(20)	NO	Hospital shift assignment.	Morning / Afternoon / Night
shift_period	NVARCHAR(10)	NO	Period within the shift.	"Early Afternoon"
is_business_hour	BIT	NO	1 if between 08:00-17:00.	1

dim_diagnosis

Type: Dimension — Static (Type 1)

Distribution: REPLICATE

Grain: One row per ICD-10 diagnosis code

Source: ICD-10 standard code table + Synthea mappings

Column	Data Type	Nullable	Description	Example
diagnosis_key	INT	NO	Surrogate primary key.	1001
icd10_code	NVARCHAR(10)	NO	ICD-10-CM diagnosis code.	"E11.9"
description	NVARCHAR(500)	NO	Full clinical description.	"Type 2 diabetes mellitus without complications"

Column	Data Type	Nullable	Description	Example
category_code	NVARCHAR(10)	YES	ICD-10 chapter code (3 chars).	"E11"
category_name	NVARCHAR(200)	YES	ICD-10 chapter name.	"Endocrine, nutritional and metabolic diseases"
body_system	NVARCHAR(100)	YES	Body system grouping.	Cardiovascular / Respiratory / Endocrine
is_chronic	BIT	NO	1 if classified as chronic condition.	1
is_communicable	BIT	NO	1 if infectious/communicable disease.	0
severity_level	NVARCHAR(20)	YES	Clinical severity classification.	Low / Medium / High / Critical

dim_payer

Type: Dimension — Type 1 (overwrite on change)

Distribution: REPLICATE

Grain: One row per insurance payer

Source: Custom Python generator (generate_claims.py)

Column	Data Type	Nullable	Description	Example
payer_key	INT	NO	Surrogate primary key.	5
payer_id	NVARCHAR(50)	NO	Natural key — payer UUID.	"pay-001"
payer_name	NVARCHAR(200)	NO	Insurance company name.	"BlueCross BlueShield"
payer_type	NVARCHAR(50)	NO	Payer classification.	private / medicaid / medicare / self-pay / other
plan_name	NVARCHAR(200)	YES	Specific plan within payer.	"Gold PPO Plan"
state_code	NVARCHAR(5)	YES	State where payer operates.	"NY"
is_government	BIT	NO	1 if government-administered payer.	0

fact_encounters

Type: Fact Table

Distribution: HASH(patient_key)

Grain: One row per patient encounter

Source: Synthea encounters.csv → Silver → Gold

Refresh: Daily batch load via ADF + Databricks

Column	Data Type	Nullable	Description	Business Rule
encounter_key	BIGINT	NO	Surrogate PK.	Auto-generated in Gold layer
encounter_id	NVARCHAR(50)	NO	Natural key from Synthea.	Unique per encounter
patient_key	INT	NO	FK → dim_patient (current version).	Must match is_current=1 row
provider_key	INT	NO	FK → dim_provider (at time of encounter).	Point-in-time lookup
date_key	INT	NO	FK → dim_date (encounter start date).	YYYYMMDD format
end_date_key	INT	YES	FK → dim_date (encounter end date).	NULL for same-day visits
payer_key	INT	NO	FK → dim_payer.	Payer at time of encounter
encounter_type	NVARCHAR(50)	NO	Type of clinical visit.	inpatient / outpatient / emergency / wellness / urgent_care
encounter_class	NVARCHAR(50)	YES	FHIR encounter class.	AMB / IMP / EMER
total_cost	DECIMAL(12,2)	YES	Total billed cost of encounter (USD).	≥ 0
payer_coverage	DECIMAL(12,2)	YES	Amount covered by insurance.	≤ total_cost
patient_cost	DECIMAL(12,2)	YES	Out-of-pocket patient responsibility.	total_cost - payer_coverage
length_of_stay_days	DECIMAL(8,2)	YES	Duration: end_date - start_date in fractional days.	0 for outpatient; > 0 for inpatient
num_conditions	SMALLINT	YES	Count of conditions diagnosed in this encounter.	≥ 0; default 0
num_medications	SMALLINT	YES	Count of medications prescribed.	≥ 0; default 0
num_procedures	SMALLINT	YES	Count of procedures performed.	≥ 0; default 0

Column	Data Type	Nullable	Description	Business Rule
readmission_flag	BIT	NO	1 if patient had prior inpatient discharge within 30 days.	Computed in Gold ETL
readmission_days	SMALLINT	YES	Days since prior discharge. Populated when readmission_flag=1.	1-30
is_emergency	BIT	NO	1 if encounter_type = 'emergency'.	Derived from encounter_type
is_inpatient	BIT	NO	1 if encounter_type = 'inpatient'.	Derived from encounter_type
etl_load_date	DATETIME2	NO	Timestamp when row was loaded.	Set by ETL process
source_system	NVARCHAR(50)	NO	Source system identifier.	"Synthea"

fact_claims

Type: Fact Table

Distribution: HASH(patient_key)

Grain: One row per insurance claim

Source: Custom Python generator → Silver → Gold

Refresh: Daily batch load

Column	Data Type	Nullable	Description	Business Rule
claim_key	BIGINT	NO	Surrogate PK.	Auto-generated
claim_id	NVARCHAR(50)	NO	Natural key.	Unique per claim
patient_key	INT	NO	FK → dim_patient.	Current version
encounter_key	BIGINT	NO	FK → fact_encounters.	Links claim to clinical event
payer_key	INT	NO	FK → dim_payer.	Payer this claim was submitted to
date_key	INT	NO	FK → dim_date (submission date).	When claim was filed
amount_billed	DECIMAL(12,2)	NO	Gross amount billed to payer.	> 0
amount_paid	DECIMAL(12,2)	YES	Amount actually reimbursed.	≥ 0; NULL if pending
amount_denied	Computed	YES	amount_billed - amount_paid.	≥ 0
adjustment_amount	DECIMAL(12,2)	YES	Contractual adjustments.	May be negative

Column	Data Type	Nullable	Description	Business Rule
processing_days	SMALLINT	YES	Days from submission to decision.	≥ 0
claim_status	NVARCHAR(30)	NO	Current lifecycle status.	submitted / approved / denied / appealed / paid
denial_reason	NVARCHAR(200)	YES	Human-readable denial reason.	"Not medically necessary"
denial_reason_code	NVARCHAR(20)	YES	Standardized denial reason code.	"CO-4", "PR-96"
is_denied	BIT	NO	1 if claim_status = 'denied'.	Derived
is_appealed	BIT	NO	1 if claim_status = 'appealed'.	Derived
is_paid	BIT	NO	1 if claim_status = 'paid'.	Derived
etl_load_date	DATETIME2	NO	ETL load timestamp.	
source_system	NVARCHAR(50)	NO	Source system.	"CustomGenerator"

fact_vitals

Type: Fact Table

Distribution: HASH(patient_key), Partitioned by date_key

Grain: One row per vital signs reading (IoT device event)

Source: IoT Simulator → Azure Event Hubs → Spark Streaming → Delta Lake

Refresh: Continuous streaming (micro-batch every 2-5 seconds)

Volume: ~500K-1M rows/day (continuous stream)

Column	Data Type	Nullable	Description	Normal Range
vital_key	BIGINT	NO	Surrogate PK.	Auto-generated (streaming)
device_id	NVARCHAR(50)	YES	IoT monitor device identifier.	"MON-A-0042"
patient_key	INT	NO	FK → dim_patient.	Current version
date_key	INT	NO	FK → dim_date.	Reading date
time_key	INT	NO	FK → dim_time.	Reading time (HHMM)
heart_rate	DECIMAL(6,2)	YES	Heart rate in beats per minute.	60-100 bpm

Column	Data Type	Nullable	Description	Normal Range
systolic_bp	DECIMAL(6,2)	YES	Systolic blood pressure (mmHg).	90-140 mmHg
diastolic_bp	DECIMAL(6,2)	YES	Diastolic blood pressure (mmHg).	60-90 mmHg
temperature	DECIMAL(5,2)	YES	Body temperature in Celsius.	36.5-37.5°C
spo2	DECIMAL(5,2)	YES	Blood oxygen saturation percentage.	95-100%
respiratory_rate	DECIMAL(5,2)	YES	Breathing rate per minute.	12-20 /min
anomaly_flag	BIT	NO	1 if any vital sign triggered alert threshold.	Set by streaming pipeline
anomaly_type	NVARCHAR(100)	YES	Type of anomaly detected.	tachycardia / bradycardia / hypertension / hypotension / fever / hypothermia / hypoxia
anomaly_severity	NVARCHAR(20)	YES	Clinical severity of anomaly.	mild / moderate / critical
window_avg_hr	DECIMAL(6,2)	YES	5-minute rolling average heart rate.	Computed by Spark Streaming
window_avg_spo2	DECIMAL(5,2)	YES	5-minute rolling average SpO2.	Computed by Spark Streaming
hr_trend	NVARCHAR(10)	YES	Heart rate trend over last 3 windows.	increasing / stable / decreasing
reading_timestamp	DATETIME2	NO	Exact timestamp of vital sign reading.	UTC timezone
etl_load_date	DATETIME2	NO	When record was written to Delta Lake.	
source_system	NVARCHAR(50)	NO	Source system.	"IoTSimulator"

Anomaly Thresholds (set in Spark Streaming pipeline):

Vital Sign	Mild Alert	Moderate Alert	Critical Alert
Heart Rate (bpm)	101-120 or 50-59	121-150 or 40-49	>150 or <40
Systolic BP (mmHg)	141-159 or 85-89	160-179 or 70-84	≥180 or <70
Temperature (°C)	37.5-38.4	38.5-39.4	≥39.5

Vital Sign	Mild Alert	Moderate Alert	Critical Alert
SpO2 (%)	93-94	90-92	<90

End of Data Dictionary — Milestone 2